

A Brain-computer Interface for Motor Imagination Based on Multi-channel EEG Analysis

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Abstract—BCI (Brain Computer Interface, BCI) system means that between the human brain and a computer or other electronic device established communication system. Motor imagination EEG signals are derived from the active consciousness of the human brain, spontaneously in the brain's movement imagination process of different parts of the limbs, without external stimulation. At present, as a mainstream brain-computer interface input signal, it has been increasingly studied. Traditional motor imagination EEG signal classification research is often based on the C3 and C4 dual-channel EEG signal features for extraction and differentiation, and the classification and recognition accuracy is low. In this paper, the wavelet transform coefficients of F3, F4, C3, C4, FZ, CZ, FC1, FC2, FC5 and FC6 multi-channel motor imagination EEG signals are introduced as features, and the extracted features are classified by Support Vector Machines (SVM). Proposed multi-channel Characteristics Analysis Method based on wavelet transform (Multi-channel Characteristics Analysis Method, MCCAM) with dual channel Analysis Method and Common Spatial Pattern (Common Spatial Pattern, CSP) Method are carried on the comparative study, found that multi-channel Analysis Method to imagine the classification accuracy of EEG signals is significantly higher than other two methods, so as to verify the effectiveness of the proposed multi-channel Method.

Keywords—Multi-channel signal analysis, Motor imagination, Wavelet transform, Common Spatial Pattern, Support Vector Machines

I. INTRODUCTION

The classification process of EEG signals usually includes sub-processes such as signal acquisition and preprocessing, feature extraction, and pattern classification. At present, the classification algorithms of motor imagination EEG signals are

mostly based on the analysis of event-related desynchronization (ERD) and event-related synchronization (ERS) signals in the sensorimotor cortex. This significant change in the EEG signal of the relevant brain area brought about by the participants' imagination of different parts of the limbs can help researchers determine the types of different motor imagination activities. The classification process usually includes the following three steps:

(1) EEG signal acquisition and preprocessing: Due to the presence of noise interference and other artifacts such as eye signals in the collected EEG signals, EEG signals should be preprocessed firstly, including filtering and pseudotrache removal process, in order to obtain relatively pure EEG signals and prepare for feature extraction of EEG signals.

(2) Feature extraction: At present, commonly used feature extraction algorithms include Common Spatial Pattern (CSP) [1], wavelet transform [2], AR model [3] and power spectrum estimation (PSD) [4]. When the wavelet transform is characterized by characterizing the local signal as a typical frequency analysis, and having a good time-frequency resolution of the frequency domain. AR model and power spectrum estimation (PSD) are based on the linear analysis process, which constructs a feature space by extracting feature parameters of the signal model. The Common Spatial Pattern is to obtain a spatial filter by calculating the diagonalization of the two types of covariance matrices to distinguish the EEG characteristics of different parts of the motion imagination. The disadvantage of the Common Spatial Pattern is that it is more sensitive to noise and computationally expensive.

(3) Pattern classification: Use the extracted feature vector of the EEG signal as the input of the classifier to design the classifier to identify the categories of different motor imaginations. Common pattern classification algorithms include ① Linear Discriminant Analysis (LDA) [5]. LDA is a

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linear learning method. The main idea is to project the two types of samples to be classified onto a straight line through a transformation matrix, and use the divergence as a measure to make the sample points of the same type cluster together, and the sample points of different types are far away from each other. ② Support vector machine (SVM) [6], SVM based on structural risk minimization principle, the construction plane is divided over the best in the feature space, as much as possible to separate the two models area. ③ Artificial neuron network [7], a classifier construction method produced by simulating the learning process of the human brain nervous system. From a theoretical point of view, independent artificial neurons are connected to form a hierarchical network. After the weights are adjusted, it can approximate any multi-input single-output nonlinear mathematical mapping.

The feature extraction process is a key process of the entire motor imagery EEG signal classification process, and its main purpose is to calculate the feature with the largest difference between different EEG signals, so as to facilitate its classification. Reference [8] uses empirical mode decomposition wavelet threshold denoising method to decompose the signal into different frequency bands, and then uses dual-channel feature wavelet packet decomposition method to extract the frequency band characteristics of C3 and C4 channels, and the resolution accuracy rate reaches 83.75%. Reference [9] uses Mallat's fast wavelet packet decomposition algorithm to extract features of C3 and C4 dual-channel EEG signals, and its resolution accuracy can reach 88.57%. As we all know, motor imagination activities are often associated with multiple brain regions. During the execution of cognitive tasks in the brain, many different areas of the brain become active, and information flows between them. It is difficult to completely separate one or two regions where activities occur (McEvoy, et al., 1997), it is far from enough to study the EEG activity of a few electrodes in a single brain area. Therefore, this study introduces the wavelet transform coefficient of F3, F4, C3, C4, FZ,CZ, FC1, FC2, FC5,FC6 multi-channel motion imagination EEG signals as features, and uses Support Vector Machines (SVM) to classify the extracted features. The proposed Multi-channel Characteristics Analysis Method (Multi-channel Characteristics Analysis Method, MCCAM) with dual channel analysis method and Common Spatial Pattern (Common Spatial Pattern, CSP) Method are carried on the comparative study, found that multi-channel analysis method to imagination the classification accuracy of EEG signals is significantly higher than other two methods, based on the potter character MCCAM left to right movement image recognition accuracy reached 98.25%. However, the recognition accuracy of the two-channel wavelet transform method is 86.67%, and the recognition accuracy of the multi-channel Common Space Pattern is 81%, thus verifying the effectiveness of MCCAM based on the wavelet change.

II. EXPERIMENTAL METHOD

A. Wavelet Transform

Signal analysis wavelet transform is the Fourier transform of a multi-scale developed in the time scale (frequency) domain for two time-frequency analysis of transient and non-stationary characteristics of the signals, signals having the

ability to characterize the local features. Due to the large amount of calculation and high complexity of continuous wavelet transform, in order to meet actual needs, discrete wavelet transform is mostly used in the signal processing process [10-11].

When $\psi(t) \in L^2(R)$ ($L^2(R)$ represents a square integrable real number space, that is, a signal space with limited energy), the value of the constant C_ψ depends on ψ , and its spectrum satisfies

$$C_\psi = \int_{-\infty}^{\infty} \frac{|\psi(w)|^2}{|w|} dw < +\infty. \quad (1)$$

Then $\psi(t)$ is called a basic wavelet or mother wavelet. After scaling and shifting the basic wavelet, we get

$$\psi_{a,b}(t) = \frac{1}{\sqrt{|a|}} \psi\left(\frac{t-b}{a}\right), a, b \in R, a \neq 0. \quad (2)$$

Call $\psi_{a,b}(t)$ a wavelet sequence. Among them: a is the expansion factor or scale factor; b is the translation factor; $1/\sqrt{|a|}$ is the normalization factor, which makes the $\psi_{a,b}(t)$ norm of $L^2(R)$ independent of the scale factor a , that is, $\|\psi_{a,b}(t)\| = \|\psi(t)\|$.

The continuous wavelet transform signal $f(t)$ is

$$W_x(a,b) = \frac{1}{\sqrt{a}} \int f(t) \psi\left(\frac{t-b}{a}\right) dt. \quad (3)$$

Refers to a discrete wavelet transform scale factor a and factor b discretization translation, rather than discrete time. The a and b in equation (3) are usually discretized by binary power series:

$$a = 2^j, b = k2^j, \quad j, k \in Z. \quad (4)$$

To get the binary wavelet (dyadic wavelet):

$$\psi_{j,k}(t) = 2^{-\frac{j}{2}} \psi(2^{-j}t - k). \quad (5)$$

The binary discrete wavelet transform of signal a is

$$c_{j,k} = \int_{-\infty}^{\infty} f(t) \psi_{j,k}(t) dt. \quad (6)$$

The db4 wavelet transform is used to decompose the C3 and C4 dual-channel EEG signals of the two types of motion imagination in three layers, and the mean value of the square wavelet coefficients of the third layer is used as the characteristic value of one channel. Figure 1 is a characteristic distribution diagram of EEG signals obtained by wavelet transform:

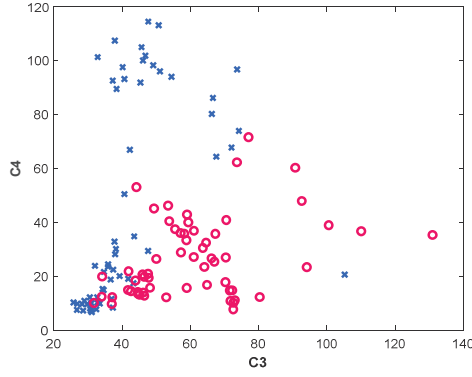


Fig. 1. Wavelet distribution of EEG characteristics.

B. Common Spatial Pattern

The specific solution algorithm of Common Spatial Pattern (CSP) [12] is as follows:

(1) Assuming that the EEG signal matrix of a single task of motor imagination is X , X is the dimension of $N \times L$, N represents the number of channels during EEG signal acquisition, and L represents the number of sampling points of each channel of each task during EEG signal acquisition. Then the covariance matrix can be obtained after normalizing X .

$$R_d = \frac{X_d X_d^T}{\text{tr}(X_d X_d^T)}. \quad (7)$$

In the formula: X_d represents the EEG number of the sample of category d , $d \in \{1, 2\}$; X_d^T represents the transposition of X_d ; $\text{tr}(\cdot)$ represents the trace of the matrix, that is, the sum of the diagonal elements of the matrix.

(2) Decompose the synthesized covariance matrix.

$$R = R_1 + R_2 = U_0 \Sigma U_0^T. \quad (8)$$

Among them, Σ is the diagonal matrix of eigenvalues, and U_0 is the corresponding eigenvector matrix. Arrange the characteristic value of Σ in descending order, and rearrange its corresponding U_0 .

(3) Find the whitening matrix, the whitening matrix P is defined as follows:

$$P = \sqrt{\Sigma^{-1}} U_0^T. \quad (9)$$

(4) Use the whitening matrix P to whiten the average normalized spatial variances R_1 and R_2 of left and right hand motor imagination respectively.

$$S_1 = P R_1 P^T, \quad S_2 = P R_2 P^T. \quad (10)$$

(5) Decomposition of principal components.

$$S_1 = U_1 \Sigma_1 U_1^T, \quad S_2 = U_2 \Sigma_2 U_2^T. \quad (11)$$

Among them, $\Sigma_1 + \Sigma_2 = I$, because the larger Σ_1 is, the smaller Σ_2 is, vice versa, so take the eigenvectors

corresponding to the largest m eigenvalues in Σ_1 and Σ_2 to form the projection matrix U , $U = (U_1' U_2')$, where U_1' and U_2' are the eigenvectors corresponding to the largest m eigenvalues. Therefore, after the whitened EEG signal is processed by U filtering, we can get The best classification feature value.

(6) Find the CSP spatial filter W .

$$W = U^T P. \quad (12)$$

Extract features from the C3 and C4 dual-channel EEG signals through the Common Spatial Pattern, take the variance of each channel of the spatially filtered data, use the logarithm of the variance of each channel as the feature value, and get two feature values for each group of data, the feature distribution of each group of data is shown in Figure 2:

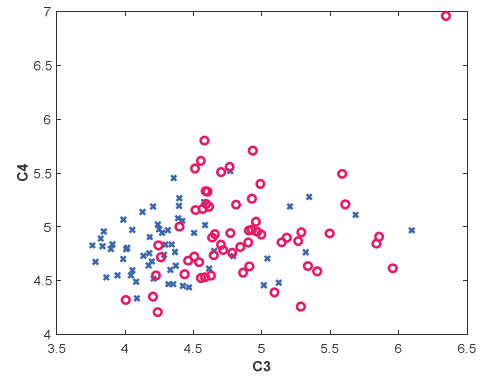


Fig. 2. EEG characteristics of common spatial pattern distribution

C. Pattern Classification Method

This paper adopts the support vector machine (SVM) [13] pattern classification method. In the classification, the linear inseparability problem is often encountered. The slack variable can be introduced into the constraint condition, and the penalty factor is introduced into the objective function to solve the problem. The optimization problem is:

$$\min \frac{1}{2} \|W^*\|^2 + C \sum_{i=1}^i x_i, i = 1, 2, \dots, N. \quad (13)$$

$$y_i ((X_i \cdot W^*) + b^*) \geq 1 - \xi_i, i = 1, \dots, N. \quad (14)$$

The slack variables ξ_i ($\xi_i > 0$), ξ_i is used to measure the distance between the actual indicated value y_i and the SVM output. Where C is the penalty factor, which controls the degree of misclassification. $k(X_i, X_j) = (\phi(X_i), \phi(X_j))$ is the mapping function, which maps the data from the original space to the high-dimensional space. Its kernel function $k(x_i, x_j)$ satisfies Mercer condition and uses radial basis kernel function [14]

$$K(x_i, x_j) = \exp(-g \|x_i - x_j\|^2), g > 0. \quad (15)$$

In the formula, the parameter g directly affects the complexity of the SVM classification algorithm. To sum up, the penalty factor C and the kernel function parameter g are the key factors that affect the SVM classification effect. Therefore, C and g can be used as optimization variables.

The results of multi-channel EEG signal pattern classification by wavelet transform and Common Spatial Pattern are shown in Table 1:

TABLE I. THE CLASSIFICATION ACCURACY OF MULTI-CHANNEL EEG SIGNAL PATTERNS

Number of channels Algorithm	2	3	5	7	10
	C3,C4	C3,C4 CZ	F3,F4,C3 C4,CZ	F3,F4,C3 C4,FZ, FC1,FC2	F3,F4,C3 C4,FZ, CZ,FC1, FC2,FC5 FC6
Wavelet transform	86.67%	88.25%	90.83%	97.33%	98.25%
Common Spatial Pattern	50.75%	52.92%	63.08%	69.75%	81%

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Data Collection

The experimental data described herein using the German company BP 32 channel EEG device for left and right hand motor imagination EEG signal acquisition. Wear an EEG cap before the start of data collection, and inject conductive paste into each electrode so that the electrode impedance will be about 5 kilohms, and the sampling frequency will be 500 Hz. The subjects were reminded through the left and right arrows in the Matlab GUI interface, and then began to imagine the movement of the left and right hands, and collected EEG data. While collecting data, mark the true type of EEG signal. The experiment process is shown in Figure 3. The display interface is "ten" in the second to third seconds, ready to start motor imagination, and the display is left or directional in the 3 to 9 seconds. The arrow symbol, the participant imagines the movement of the left and right hands according to the arrow tips. Each experiment carried out 20 times of motor imagination, collected 6 sets of data, and got a total of 120 sets of motor imagination EEG data, half of the data is used as the training set for feature extraction and classifier training, and the other half is used as the test set to test the classification accuracy.

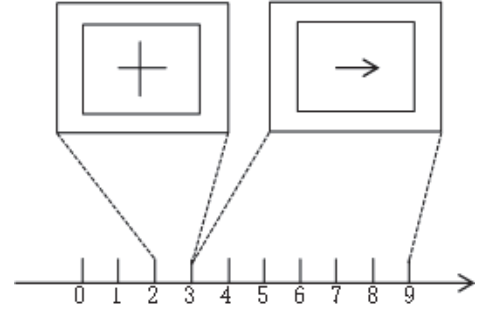
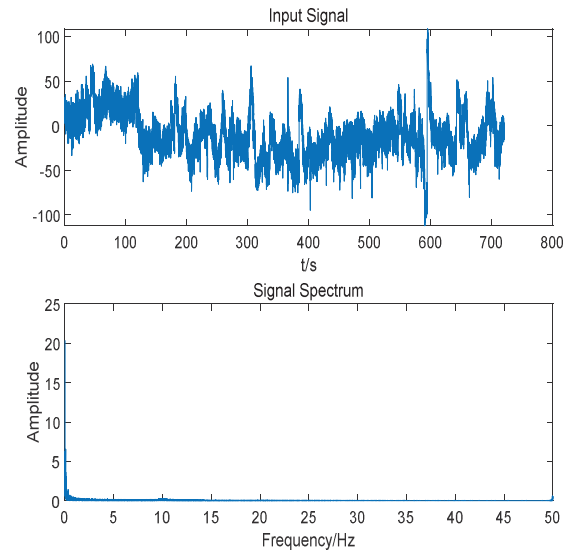


Fig. 3. Experimental sequence diagram

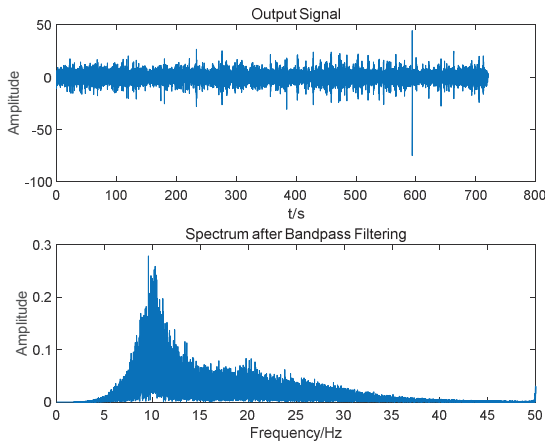
B. Pretreatment

To filter and de-artifact preprocess the collected EEG signals, the commonly used methods are: Butterworth filter [15], independent component analysis (ICA) [16] and principal component analysis (PCA) [17]. Principal component analysis (PCA) is a classic method used to estimate the parameters of a linear model. The ICA method decomposes the EEG signals into independent components based on the distribution of each lead of the EEG signal, then removes the noise and artifact components, and finally reconstructs the EEG signal to achieve the effect of removing noise and artifacts. This method makes the separated independent components approximate to each source signal to the greatest extent, that is, establishes the objective function to find the optimization to realize the decomposition of the independent components. Theoretically, it is generally believed that the interfering signals such as ocular electricity and myoelectricity in the EEG signal are all generated by independent sources, so the ICA method can be used to remove artifacts. This study mainly used ICA and Butterworth filter to preprocess the collected motor imagination EEG signals.

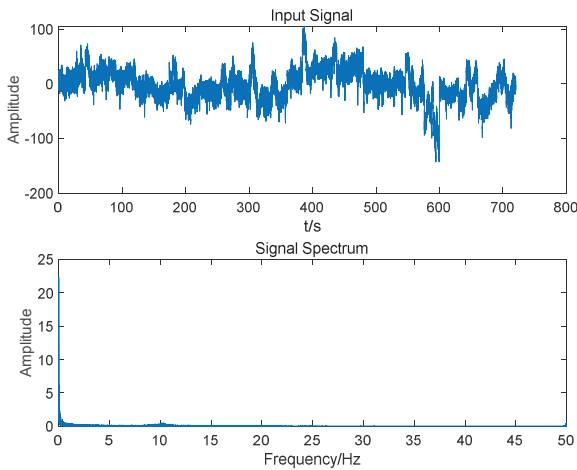
This article uses a total of 10 EEG signal channels. Figures 4 and 5 show the time-frequency domain diagrams of the EEG signals before and after the C3 and C4 channels.



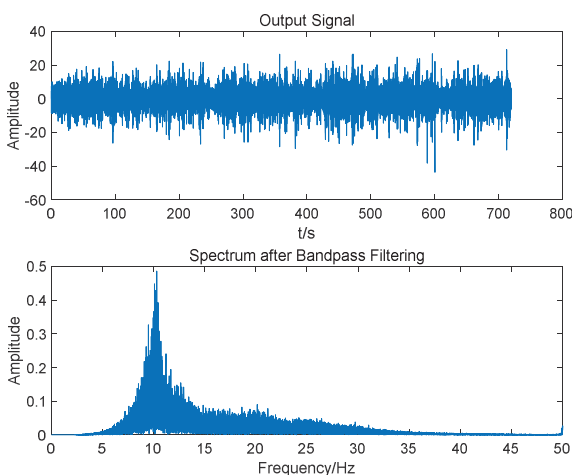
(a) Time-frequency domain diagram of EEG signal before C3 channel filtering



(b) Time-frequency domain diagram of EEG signal after C3 channel filtering
Fig. 4. Time-frequency domain diagram of C3 channel before and after filtering



(a) Time-frequency domain diagram of EEG signal before C4 channel filtering



(b) Time-frequency domain diagram of EEG signal after C4 channel filtering
Fig. 5. Time-frequency domain diagram of C4 channel before and after filtering

C. Experimental Results

First, for the traditional C3 and C4 dual-channel EEG signals, two methods of wavelet transform and Common Spatial Pattern are used to design the classifier and test the classification effect. The classification accuracy rate is shown in Table 1, only 86.67% and 50.75%. Incorporating the third channel CZ further, the accuracy rate rose to 88.25% and 52.92%. Gradually add more signals F3, F4, FZ, FC1, FC2 of the same brain area channel, and the accuracy of FC5 and FC6 tests will increase accordingly. When 10 channels of EEG signals of F3, F4, C3, C4, FZ, CZ, FC1, FC2, FC5, FC6 are used for feature extraction and classification, the classification accuracy rate reaches 98.25% and 81%. The experimental results show that the use of multi-channel feature extraction method can significantly improve the classification accuracy of motor imagination EEG signals, and feature extraction based on wavelet multi-channel analysis is better than the effect of Common Spatial Pattern classification method based on multi-channel feature extraction.

IV. CONCLUSIONS

This study uses the physiological characteristics of the motor imagination EEG signals, selects the C3, C4 channels and other multi-channel EEG signals related to the sensory motor cortex area for analysis, and uses the discrete wavelet transform to extract the features of the left and right hand motor imagination EEG signals. The Support Vector Machines (SVM) classifies the extracted features and tests the accuracy of the classification. The experimental test results show that the classification effect of the feature extraction method based on multi-channel signals is significantly better than that of the feature extraction method based on traditional dual-channel signals. The wavelet analysis method based on multi-channel feature extraction is better than the dual-channel analysis method based on multi-channel feature extraction and the Common Spatial Pattern method. This conclusion further verifies and reveals the physiological phenomena of motor imagination activities associated with multiple sub-areas in the sensory motor brain area from the perspective of brain-computer interface experiments. It can be used to optimize the design process of the brain-computer interface for motor imagination.

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